

RAG Ingestion Incident Postmortem / 文档解析复盘

Demo corpus generated for RAG testing. Version: 2026-06-08. Source: synthetic document.

1. Incident Summary

On 2026-06-05, the document ingestion service returned incomplete chunks for several Chinese PDF files. The issue affected retrieval quality but did not affect authentication, billing, or user data integrity.

The immediate cause was an overly aggressive fixed-size splitter. It cut sentences in the middle and allowed table-of-contents fragments to enter the vector index.

2. Impact

Users could upload files successfully, but some answers cited low-quality chunks. In several cases, the top retrieval result was a table of contents entry instead of the paragraph containing the actual answer.

The problem was most visible in procurement documents, policy manuals, and reports that contain page headers, footers, long dot leaders, and multi-page tables.

3. Root Cause

The parser extracted text correctly, but the chunking strategy did not use semantic boundaries. It ignored headings, paragraph breaks, Chinese punctuation, and table structure. Metadata was partially retained but did not include section names.

The eval suite only checked whether ingestion completed. It did not include retrieval-level tests such as top-1 relevance, top-3 contains answer, source page accuracy, or citation quality.

4. Corrective Actions

Add a chunk quality filter that removes very short chunks, chunks dominated by punctuation, and likely table-of-contents fragments. Add sentence-aware splitting for Chinese documents using punctuation such as 。 ！ ？ ； as preferred cut points.

Add retrieval eval cases with exact-match queries, paraphrase queries, and cross-document comparison queries. Each eval case should record query, expected_source, expected_page, top_k_results, and pass/fail.

5. Follow-up Metrics

Key metrics include parse_success_rate, low_quality_chunk_ratio, top1_relevance, top3_contains_answer, citation_page_accuracy, average_latency_ms, and embedding_cost_per_document.

A weekly regression report should compare changes in parser, splitter, embedding model, and reranker settings.

Table 1: Retrieval Quality Checks

Check	Expected	Failure example
Chinese encoding	Readable Chinese text	□□□ or mojibake symbols

Sentence boundary	No hard cut in middle of sentence	根据模 / 型细分
Metadata	source/page/chunk_id retained	answer without source
Table handling	Markdown or structured rows	columns flattened into random text

Synthetic postmortem for retrieval evaluation. Designed to test English + Chinese mixed document parsing.